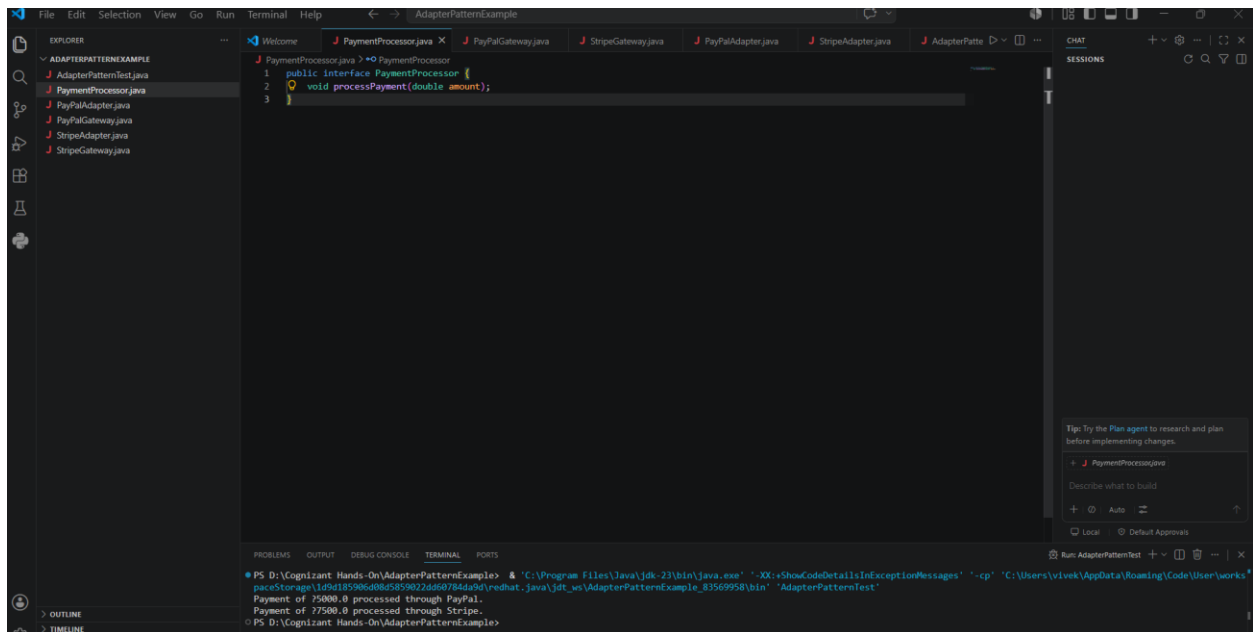


Hands-On

Exercise 4: Implementing the Adapter Pattern

In this module I have successfully implemented, executed and tested the Adapter Pattern in java by creating a project called AdapterPatternExample. I've developed a PaymentProcessor interface that would be the common target interface and created several payment gateway classes that each have their own methods and interfaces. I created adapter classes that implement the PaymentProcessor interface and convert payments into calls to the gateway's methods to seamlessly integrate these gateways. Once implemented, I executed the program and tried various payment gateways using their respective adapters to ensure that they can all be interchanged using a common interface. This practical exercise helped me appreciate the Adapter Pattern as a way to make incompatible interfaces compatible, to get code reusability and to integrate third party services into an application.

Coding:

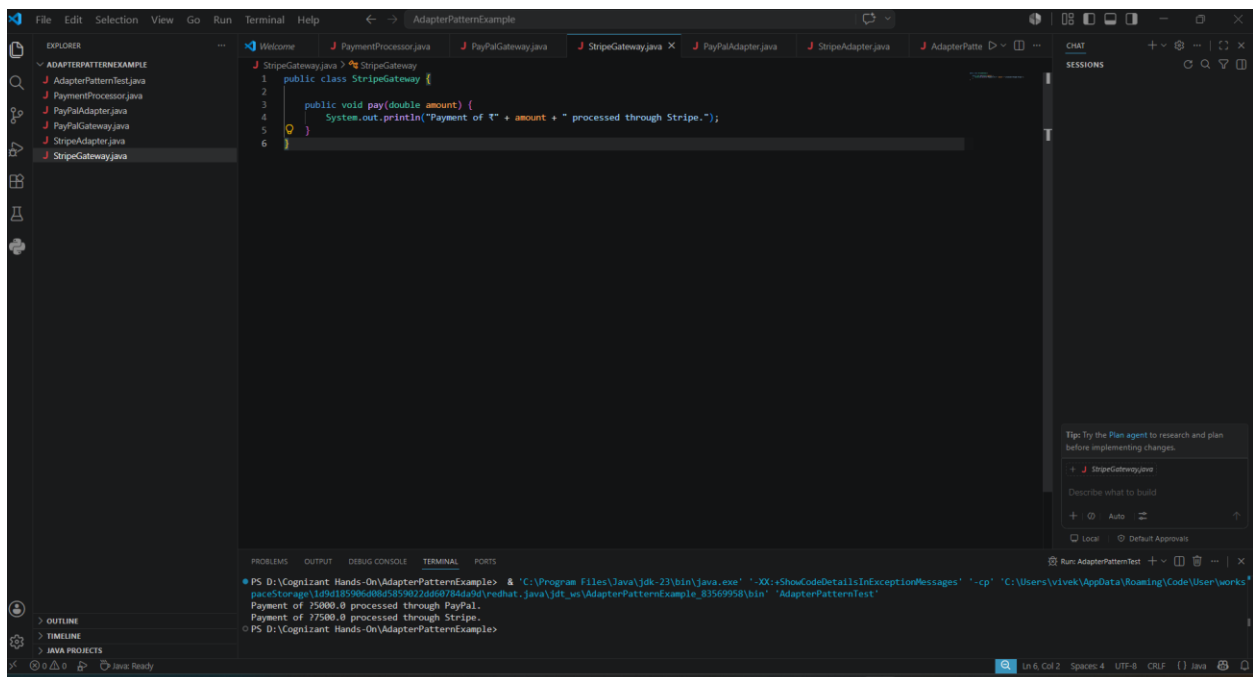
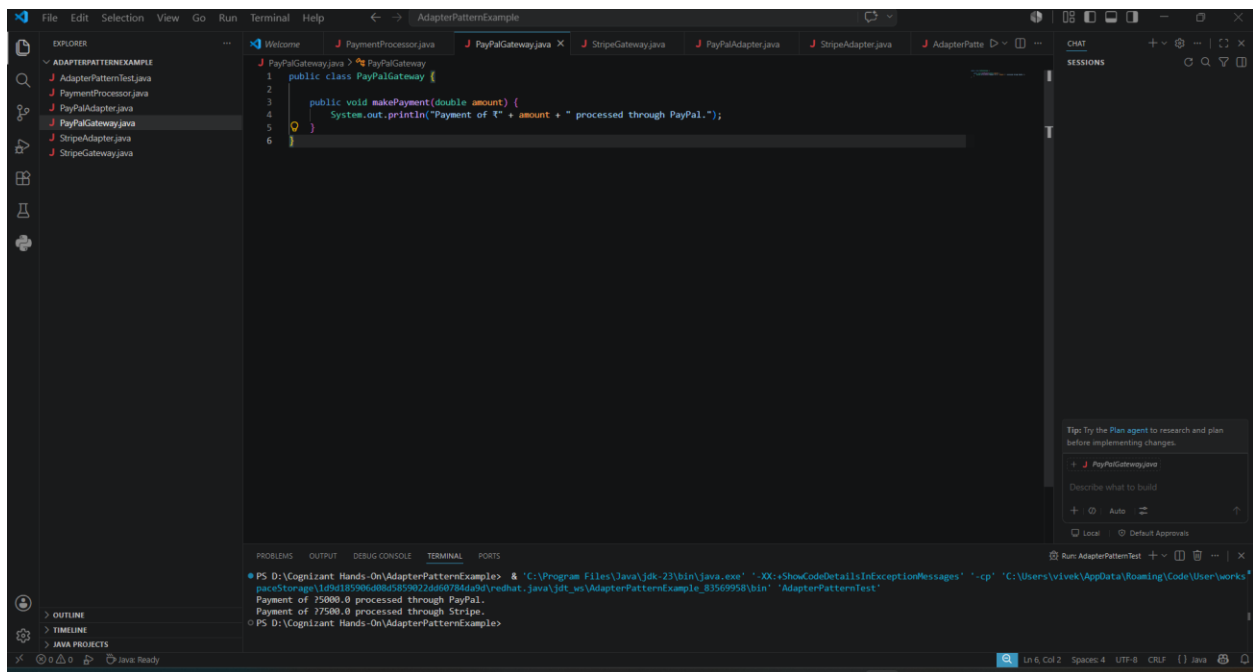


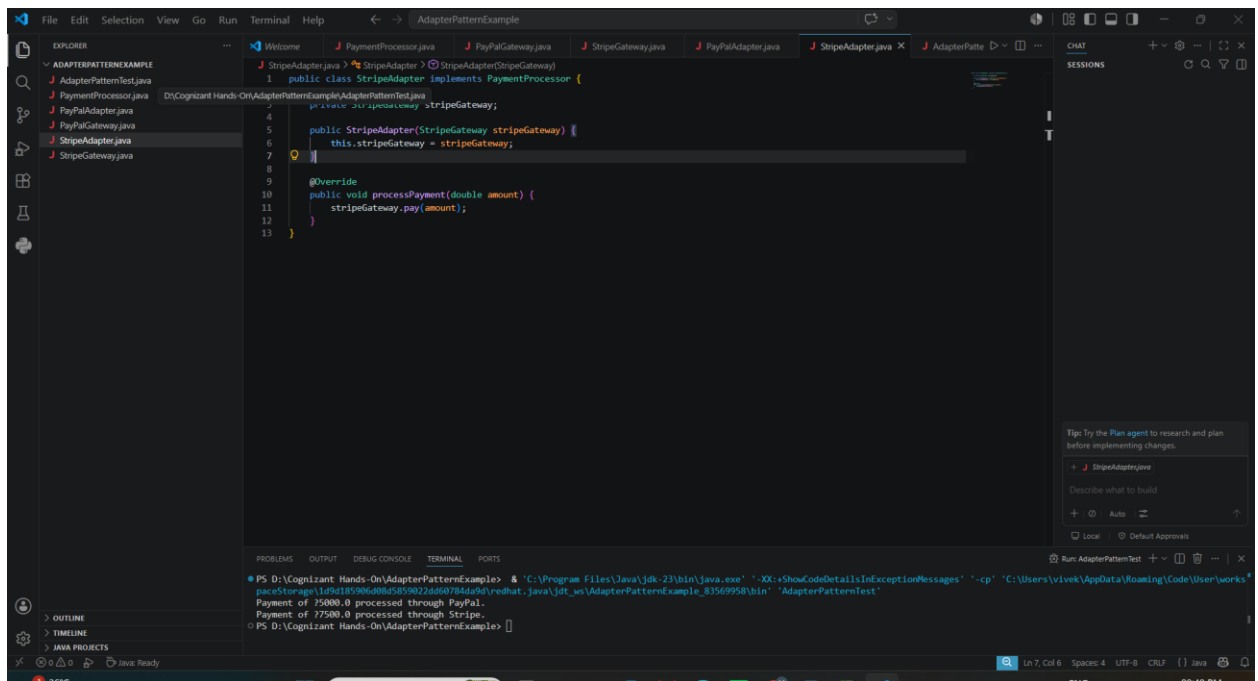
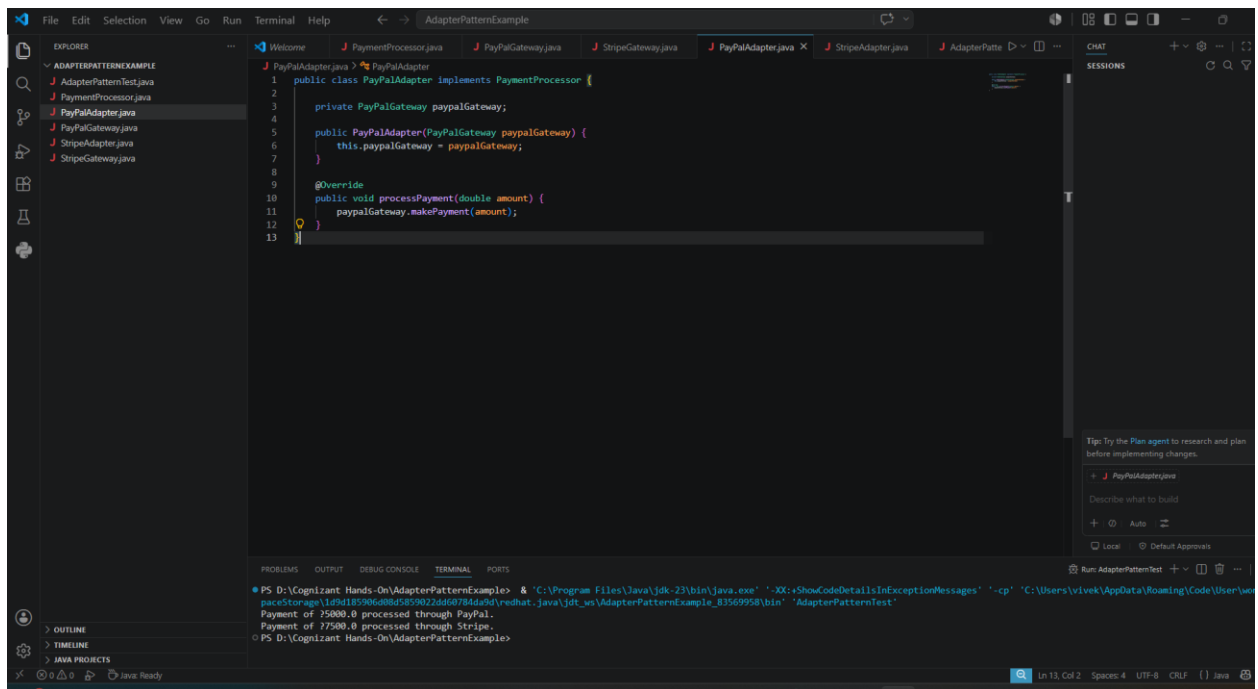
The screenshot displays an IDE window titled 'AdapterPatternExample'. The Explorer panel on the left shows the project structure with files: AdapterPatternTest.java, PaymentProcessor.java, PayPalGateway.java, StripeGateway.java, PayPalAdapter.java, StripeAdapter.java, and AdapterFatte.java. The main editor shows the PaymentProcessor.java file with the following code:

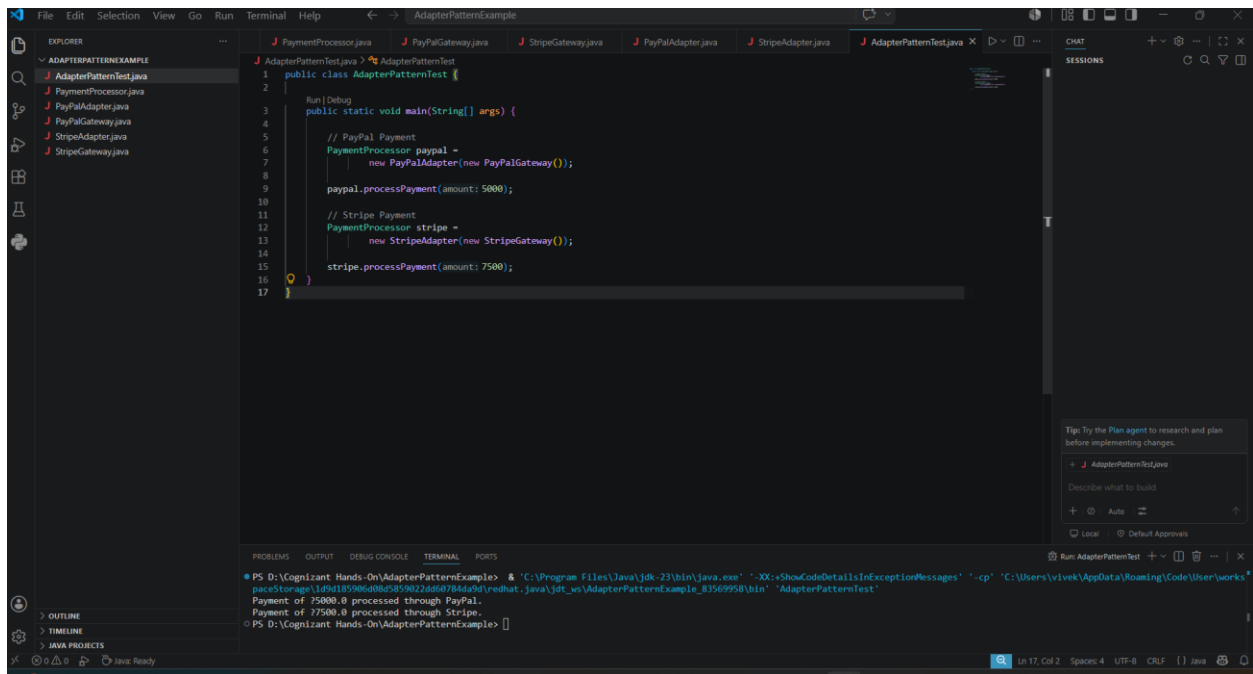
```
1 public interface PaymentProcessor {  
2     void processPayment(double amount);  
3 }
```

The bottom panel shows the terminal output with the following text:

```
PS D:\Cognizant_Hands-On\AdapterPatternExample> java -cp 'C:\Program Files\Java\jdk-23\bin\java.exe' -XX:+ShowCodeDetailsInExceptionMessages -cp 'C:\Users\vivek\AppData\Roaming\Code\User\workspaceStorage\1f9d18f596e0b0555022690784d4f4e\redhat.java\jdk_ws\AdapterPatternExample_83569958\bin' AdapterPatternTest  
Payment of 75000.0 processed through PayPal.  
Payment of 77500.0 processed through Stripe.  
PS D:\Cognizant_Hands-On\AdapterPatternExample>
```







Output:

